

Measuring error

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Take-aways from last class activity

- ▶ Even if the model is correctly specified, prediction performance on the **training** data is a biased estimate of prediction performance on **new** data
- ▶ The bias gets worse as the number of explanatory variables in the model increases!

Bias of the training error

Let $\mathcal{T} = \{(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)\}$ be a training set, and let $\hat{h}$ be a model fit on $\mathcal{T}$ by maximizing the likelihood.

Two big results (under regularity conditions):

Deriving bias of the training error

Let $\mathbf{y} = (Y_1, \dots, Y_n) \stackrel{iid}{\sim} g$ for some unknown distribution g be our **training set**, and let $\mathbf{y}^* = (Y_1^*, \dots, Y_n^*) \stackrel{iid}{\sim} g$ be another sample from g , *independent* of $\mathbf{y}$.

Error quantities

Let $\mathbf{y} = (Y_1, \dots, Y_n) \stackrel{iid}{\sim} g$ for some unknown distribution g be our **training set**, and let $\mathbf{y}^* = (Y_1^*, \dots, Y_n^*) \stackrel{iid}{\sim} g$ be another sample from g , *independent* of $\mathbf{y}$.

Idea

$$\mathcal{R}_{\text{train}}(\hat{\boldsymbol{\theta}}) = -\frac{2}{n} \log f_{\hat{\boldsymbol{\theta}}}(\mathbf{y})$$

$$\mathcal{R}_{\mathcal{T}}(\hat{\boldsymbol{\theta}}) = \mathbb{E}_{\mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\boldsymbol{\theta}}}(\mathbf{y}^*) \middle| \mathbf{y} \right]$$

$$\mathcal{R} = \mathbb{E}_{\mathbf{y}}[\mathcal{R}_{\mathcal{T}}(\hat{\boldsymbol{\theta}})] = \mathbb{E}_{\mathbf{y}, \mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\boldsymbol{\theta}}}(\mathbf{y}^*) \right]$$

Lemma 1 proof

AIC

Let $\hat{h}$ be a model fit on training data $\mathcal{T}$ by maximizing likelihood (equivalently, minimizing deviance). Let p be the number of parameters in the model (i.e. for logistic regression, $\beta \in \mathbb{R}^p$).

Example

```
m_dengue <- glm(Dengue ~ Age + WBC + PLT,  
                data = dengue, family = binomial)  
m_dengue$deviance
```

```
## [1] 5200.823
```

```
m_dengue$aic
```

```
## [1] 5208.823
```